

Wheel sizing

min speed 0.15km

max speed 0.25km/h

min: 1 kg 5 days

200g/day

max: 1kg/hour

?

2cm

5cm

Min particle size 2cm, max 5cm

Density 1.5g/cm³

8 Wheels

8 wheels

0.07 ~~ms⁻¹~~ max

~~0.04 min~~

0.0416 min

6 wheels

min

80mm

max

130mm

Drive system

max 10RPM

for 600m/day

Speed requirement

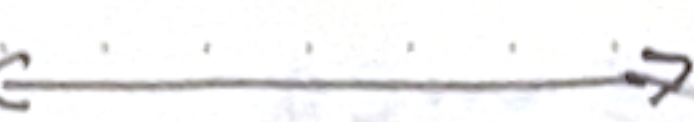
= 0.001ms⁻¹

24 x 60 x 60 = 86400 s/day

600 / 86400 = 0.001

Home

30am



Regolith



600meters per collection cycle

$7 \times 10^{-3} \text{ ms}^{-1}$

Wheel sizes

Smallest if feasible

$$\pi (80 \times 10^{-3}) \rightarrow 251 \text{ mm}$$

$$80 \text{ mm} \times \pi = \text{circumference} = 251 \text{ mm}$$

m/min? ?

min wheel size

Would save wear but not ideal

be so close to min requirements

$$251 \times 10^{-3} \times 10?$$

$$0.251 \times 10 \text{ m/s} \div 60$$

$$= 0.04186$$

Smallest just feasible

$$0.04 > 0.0416$$

$$0.042 : \theta - m$$

min allowed speed

$$0.0416 \times 60 = \text{m/minute}$$

$$70 \text{ } 0.25 \text{ m circumference}$$

$$0.25/\text{m}$$

$$.160 \text{ km/h}$$

80 mm = Not Feasible.

85 mm = Feasible.

$$85 \times \pi = 267.0 \text{ mm}$$

$$85 \times \pi = 267.0 \text{ mm} \times 10 = \frac{2670}{1000} \text{ m/s}$$

$$= \frac{2.670}{60}$$

$$= 0.0445 \text{ m/s}$$

Feasible.

$$0.160 \text{ km/h} \approx 70.15$$

Wheel size = 85 mm diameter



$$d = 85 \text{ mm}$$

Key is 5 days.

5 trips of 200g each.

74m for 220 per day.

but container volume safety factor at 2x.

least amount of trips optimal.

end of

Wheels section

Container volume

Rocks don't work perfectly.

$$\frac{220}{1.5} = 146.7$$

x2

$$= 293$$

$$\approx 300 \text{ cm}^3$$

← container size.

x2 Volume is reasonable to account for this.

Aim for big rocks

~~Aim~~ Pros

- less work done to collect

cons

- Bad packing

- Aiming for big rocks ~~the~~ increases chance of oversized particles.

Conclusion: Aim for ~~mid~~ sized, leaning towards smaller rocks, mostly for increased packing capacity.

4cm diameter roughly spherical ~~of~~ regolith samples are approximately 50g.

9cm ~~Q~~

$$\frac{4}{3}\pi r^3 = \frac{4}{3}\pi (2.0)^3 = 33.51 \text{ cm}^3$$

$$m = 33.51 \times 1.5 = 50.27 \text{ g}$$

max 220g/day.

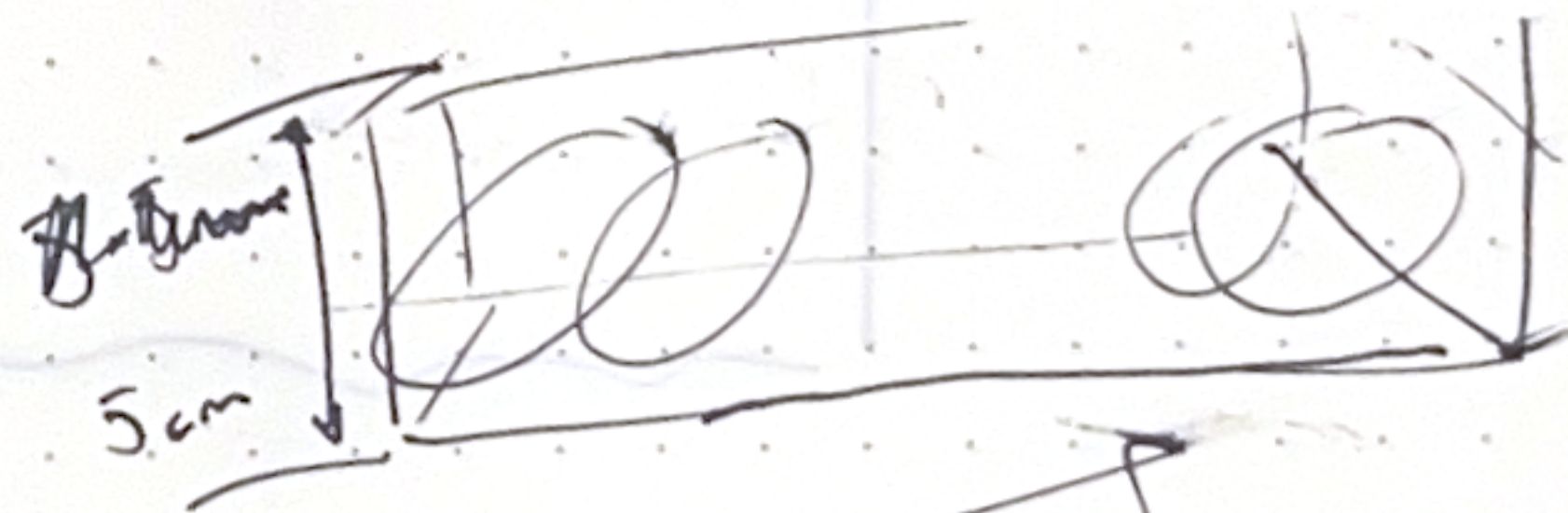
4 rocks $\approx$ 4cm diameter

18 rocks $\approx$ 2.5cm diameter.

container shape

less dead space w/ more small rocks.

~~Assume~~ Assume ~~spherical~~ spherical rocks are cubes as worst case packing scenario.



18g

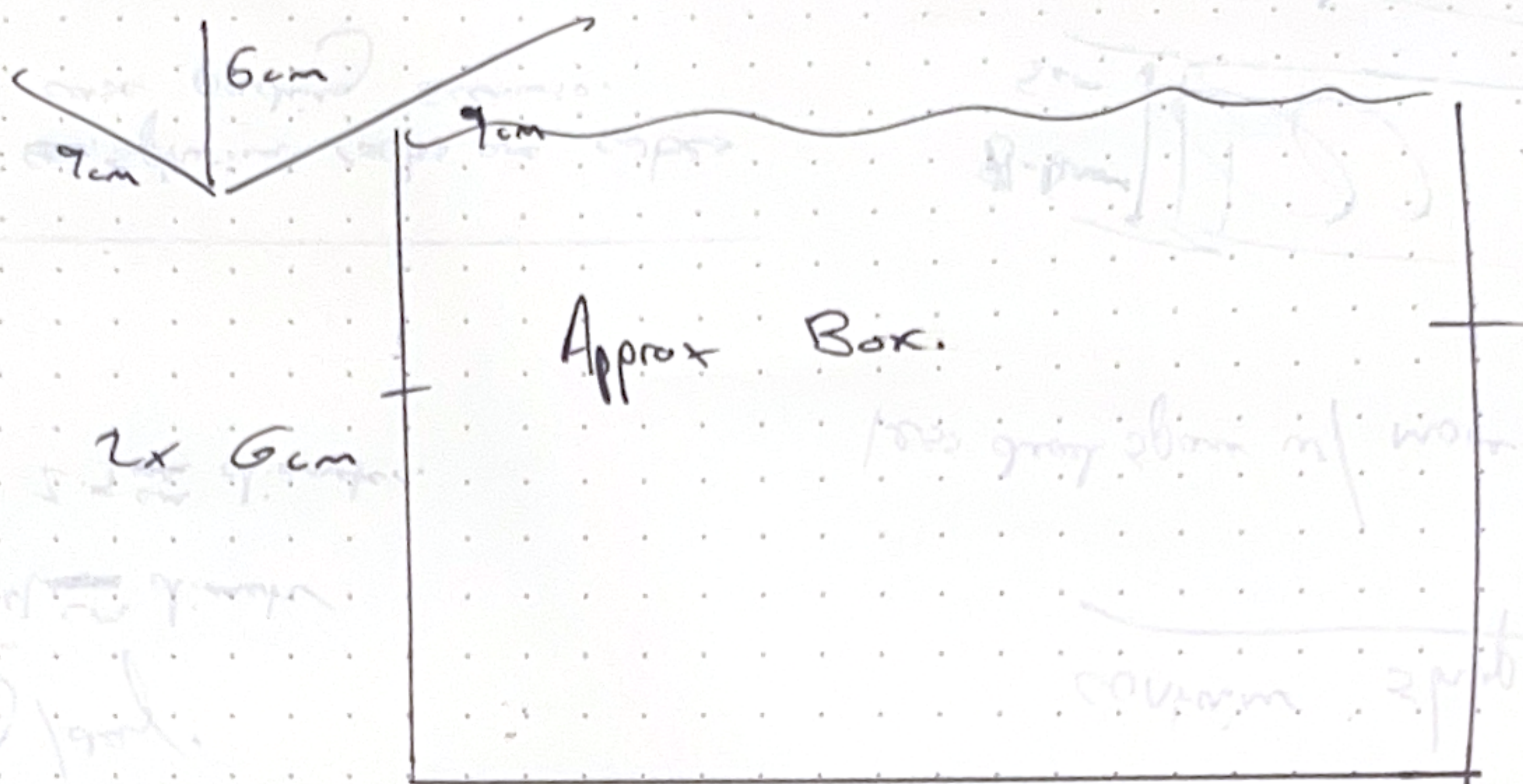
if we got 4

3 rocks high
3 rocks high

= 18 rocks

Worst rock in selection is $3 \times 3 \times 3$.

Therefore container should be able to fit 18 ($3 \times 3 \times 3$) rocks.



$$W = 23.21 \times 1.2 = 20.5 \text{ 9cm}$$

$$\frac{3}{4} \times 1.2 = \frac{3}{4} \times (5.0) = 3.75 \text{ 10cm}$$

Decided on 1 trip per day w/ safety allowance being put on rock collection ~~amount~~ amount as there is more variation in rock

Aim for between 2 & 3cm diameter rocks, with an average of 2.5cm ^{roughly spherical}

Power pack size

6 motors 3 watts

(unar daylight period 354 hours)

$$6372 \text{ Wh} = 25 \text{ kg battery}$$

28 day moon cycle

All of moon isn't always lit.

Moon light cycle.

600m/day + 100m for moving around site.

700m/day

$$700 / 0.0445$$

$$= 15,730s / 60 / 60$$

= 4.37 hours of ~~wheel spinning~~ motor usage.

$$4.37 \times 3(6) = 78.7Wh$$

0.147 litres of cells

$$140.7 cm^3$$

1 cell

Single cell volume: $0.9^2 \times \pi \times 6.5 = 16.54 cm^3 \rightarrow = 8.5 \text{ cells}$

Solar panel recharges

370g.

$$20W \times 24hrs = 480Wh/day \approx 80 \text{ required.}$$

Allow 10 cells for extra power usage by other components on rover.

